

- By Theorem 3, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.
- By the row reduction algorithm and Theorem 1, the augmented matrix can be converted to a unique rref form.
- By Theorem 2, the matrix A has at least one free variable.
- $\therefore$ There exists at least one column having no pivots.

Case 1: Exactly one column has pivot.
Conditions for a matrix to be in echelon form

- All non-zero rows must be above the zero rows.
 - For each row the leading entry must be in a column to the right of the columns of the leading entries of all rows above it.
 - For each column having a pivot, all positions below it must contain 0.
- Suppose, Row 1 has the pivot in Column 1. $\therefore$ Row 1 Col 1 = $\square$
 From Cond'n 2, Row 2 Col 1 = 0. Column 2 has no leading entries. $\therefore$ Row 2 Col 2 = 0. Row 1 Col 2 = * ($\because$ there is no constraint upon it)

Final echelon form of $A = \begin{bmatrix} \square & * \\ 0 & 0 \end{bmatrix}$

- Suppose, Row 1 has the pivot in Column 2. $\therefore$ Row 1 Col 2 = $\square$
 From Cond'n 3, Row 2 Col 2 = 0. Col 1 must have no pivots.
 $\therefore$ Row 1 Col 1 = Row 2 Col 1 = 0.

Final echelon form of $A = \begin{bmatrix} 0 & \square \\ 0 & 0 \end{bmatrix}$

Case 2: No column have pivots.

From the defn. of pivots, the final echelon form of $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \square$

25) A is a 4×2 matrix, $A = [a_1 \ a_2]$, and a_2 is not a multiple of a_1 .

Solution: Since a_2 is not a multiple of a_1 , $\therefore a_1$ and a_2 are linearly independent.
 $\therefore$ The matrix equation $Ax=0$ has a unique trivial solution.